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Wild, tamed, and domesticated: Three fire macroregimes for global pyrogeography in the Anthropocene

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Abstract

Climate and natural vegetation dynamics are key drivers of global vegetation fire, but anthropogenic burning now prevails over vast areas of the planet. Fire regime classification and mapping may contribute towards improved understanding of relationships between those fire drivers. We used 15 years of daily active fire data from the MODIS fire product (MCD14ML, collection 6) to create global maps of six fire descriptors (incidence, size inequality, season length, interannual variability, intensity, and fire season modality). Using multiple correspondence analysis (MCA) and hierarchical agglomerative clustering, we identified three fire macroregimes: Wild, Tamed, and Domesticated, each of which splitting into prototypical and transitional regimes. Interpretation of the six fire regimes in terms of their main drivers relied on the global maps of anthromes and Köppen climate types. The analysis yielded a two-dimensional space where the principal dimension of variability is primarily defined by interannual variability in fire activity and fire season length, and the secondary axis is based mainly on fire incidence. The Wild fire macroregime occurs mostly in cold wildlands, where burning is sporadic and fire seasons are short. Tamed fires predominate in seasonally dry tropical rangelands and croplands with high fire incidence. Domesticated fires are characteristic of humid, warm temperate and tropical croplands and villages with low fire incidence. The Tamed and Domesticated fire macroregimes, representing managed burning, account for 86% of all active fires in our dataset and for 70% of the global burnable area. Fourteen percent of active fires were found in the cold wildlands, and in the rangelands and forests of steppe and desert climates of the Wild macroregime. These results highlight the extent of human control over global pyrogeography in the Anthropocene.

KEYWORDS
active fires, anthromes, climate, global fire regimes, hierarchical clustering, multiple correspondence analysis

INTRODUCTION

Human societies have modified the global environment to such an extent that Crutzen (2002) proposed the

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Anthropocene as a new geological era on planet Earth, purportedly starting in the late 18th century and characterized by fast growth in the use of fossil fuels, and rapid increase in the atmospheric concentration of carbon dioxide (Steffen et al., 2007). It has thus become impossible to understand or manage most ecosystems without considering human activity, which is now a major geological force, comparable to that of the climate system (Ellis, 2015). Fire has vastly increased the ability of humans to modify their environment since at least 200,000 years ago (Glikson & Groves, 2015; Scott et al., 2013), with the transition from the natural, pre-human pyric phase to hunter-gatherer fire regimes, and even further with the subsequent emergence of agricultural and industrial fire regimes (Bowman, 2015; Bowman et al., 2011; Pyne, 2001).

Although the biophysical environment remains a key element of global fire activity, human agency became increasingly important, and anthropogenic vegetation burning now prevails over vast areas of the planet (Bowman et al., 2020; Lauk & Erb, 2009; O'Connor et al., 2011; Pereira et al., 2019). Over 40 years ago, when Crutzen et al. (1979) first alerted to the global scope, magnitude, and potential environmental impacts of biomass burning, they already observed that it was mostly anthropogenic. Andreae (1993) identified a series of motivations for vegetation burning, including forest and shrubland clearing for agricultural expansion; control of shrub and weed invasion of croplands and pastures; promoting nutrient recycling on pastures and croplands; forest fuel management; and vegetation clearing for hunting and gathering.

During the 1990s, land use change processes such as tropical deforestation and savanna conversion to pasture and cropland, combined with global warming, suggested a growing trend in area affected by both anthropogenic fire and wildfire, with increasingly severe environmental impacts (Crutzen & Andreae, 1990; Goldammer & Price, 1998). However, both burned area (Andela et al., 2017) and pyrogenic trace gas emissions (Knorr et al., 2016) have decreased since the late 20th century, mainly due to land use changes, and despite a substantial lengthening of the fire season throughout vast areas of the planet (Jolly et al., 2015). This unforeseen result of the interplay between human and climatic factors highlights the complexity of relationships between drivers of global fire activity trends. The development of pyrogeography, the study of the spatial and temporal distribution of fire under the influence of climate, vegetation, and human activity (Scott et al., 2013), which includes fire regime mapping, may contribute to improved understanding of such relationships (Bowman et al., 2009).

The definition of fire regime incorporates a core group of parameters describing which fires occur when and where, according to frequency, size, seasonality, intensity, and type (sensu stricto definition). It may also include antecedent variables, which represent enabling circumstances, or prerequisites for fire ignition and spread (e.g., climate, population, land use), and its immediate consequences, characterized by the various impacts of fire (sensu lato definition; Krebs et al., 2010). Chuvieco et al. (2008) and Archibald et al. (2013) developed classifications and maps of global fire regimes, using descriptors such as active fire density, interannual variability, fire season duration, fire size, fire intensity, and extent of area burned. All of these fire characteristics are prone to change due to anthropogenic modification of land cover and of spatiotemporal ignition patterns (Archibald, 2016; Bowman et al., 2011).

In the present study, we propose to develop a hierarchical sensu stricto global fire regime classification and interpret it in terms of climatic and anthropic antecedent variables. We also assess the relative importance and global distribution of managed fire regimes and wildfire regimes. The proposed fire regime classification is based on data from the early 21st century, although the main fire activity patterns and fire–environment relationships it unveils are likely to have been evolving at least since the "Great Acceleration" of the mid-20th century (Steffen et al., 2007), which Zalasiewicz et al. (2015) proposed as the stratigraphically optimal boundary to mark the beginning of the Anthropocene.

DATA AND METHODS

Data

We used 15 years of daily active fire data from the MODIS fire Product (MCD14ML collection 6, https://earthdata.nasa.gov/earth-observation-data/near-real-time/firms/mcd14ml) (doi: 10.5067/FIRMS/MODIS/MCD14ML) spanning the period 1 March 2003 to 28 February 2018, with a total of 66,524,069 fire counts. This dataset reports the location and timing of active fires at native resolution (1 km at nadir) for the MODIS Terra and Aqua satellites, and were aggregated to 1° spatial resolution. Since the number of grid points per unit of area increases with latitude, a correction on the MCA was performed using a procedure analogous to the geographical data variance equalization of Wilks (2019), to compensate for the oversampling of high latitude features. "*Multiple correspondence analysis (MCA)*" provides additional details concerning this procedure.

We chose to rely solely on active fire data, which are capable of detecting the small burns typical of intensively managed areas. Giglio et al. (2018) reported that MODIS

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burned area products often miss small (≲100 ha) fire scars, leading to severe underestimation of area burned in croplands, a feature that subsists in the latest product (MCD64A1, collection 6). Roteta et al. (2019) and Ramo et al. (2021) revealed that area burned in Africa, estimated using 20-m spatial resolution Sentinel-2 data, is 80% larger than the area burned mapped in MCD64A1, collection 6. They mention that the main differences between the two data sets were found in regions where small fires (≲100 ha) account for a significant proportion of total area burned. Archibald et al. (2013) faced this problem in their pyromes RCS (Rare-Cool-Small) and ICS (Intermediate-Cool-Small, the most anthropogenic pyrome), where they had trouble detecting area burned, but not active fires. Chen et al. (2017) referred a similar issue in their development of a fire regimes map for China. Therefore, we considered that the inability of global burned area products to detect most of the area burned in croplands and other small fires (e.g., deforestation burns) would bias the analysis, leading to underestimation of the importance of predominantly anthropogenic fire regimes. Active fire data have the ability to capture small, even subpixel fires and thus are much less susceptible to this kind of bias. Evidently, burned area data allows for the construction of widely used fire regime descriptors, such as fire cycle and fire return interval. However, even the entire time series of data available from MODIS is too short for reliable quantification of these descriptors in regimes with infrequent burning, such as boreal forests and deserts, which is another reason why we chose not to use them in our analysis.

Since we did not use burned area data, we had to derive a novel variable to characterize the size of fire events, represented by spatiotemporal clusters of active fires. First, active fire events were individuated according to the procedure developed by Oom et al. (2016) and then their size distribution inside each 1° cell was characterized with the Gini index (Gini, 1936). We also introduced fire season modality as a new variable sensitive to anthropogenic influence on the timing of vegetation burning, as defined by Benali et al. (2017).

Variables for fire regime characterization

We developed five quantitative variables (incidence, size inequality, season length, interannual variability, and intensity) and one qualitative variable (season modality) to perform the sensu stricto characterization of fire regimes. These variables represent spatial (fire incidence and fire size distribution) and temporal (interannual variability, season length, and season modality) patterns of fire activity, and a key fire characteristic (fire intensity). Two qualitative variables (anthromes and Köppen climate classes) assist with fire regime interpretation, representing key antecedent conditions and broadening the scope of the analysis to the sensu lato definition of Krebs et al. (2010).

Figure 1a–h displays global maps for the eight categorized variables. A brief description of the variables follows.

Incidence, IC (fire density)

Incidence is defined as the total number of screened MODIS active fire counts per 1000 km^2 for the entire study period, and is a basic metric of fire activity (Chuvieco et al., 2008; Dwyer et al., 2000; Krawchuk et al., 2009). Only cells with more than 10 fire counts over the 15 years were analyzed (Figure 1a). To avoid multiple counting at high latitudes of fire observations falling in the same 1-km^2 grid cell, only one fire per day was considered.

Size inequality, SI (dimensionless)

We used the Gini coefficient (Gini, 1936) to characterize the size distribution of active fire spatiotemporal clusters. Each fire cluster was identified using a modified version of the Oom et al. (2016) fire individuation algorithm. Grid cells where active fires are concentrated in a small number of large clusters display high values of the Gini coefficient, while cells where active fires are distributed through a large number of small clusters have low values. We used a 2-day time-gap threshold (Oom et al., 2016) to calculate the size distributions. Spatial–temporal aggregation of active fires is stopped at the edges of each 1° cell, and thus no fire event extends across a cell edge. The final Gini coefficient value assigned to each cell is the overall value for the 15 years (Figure 1b). The segmented categories range from "very even," where the total number of active fires in the cell is dispersed over several clusters of similar size, typically small, to "very uneven," where the total number of active fires in the cell is concentrated in a small number of very large clusters.

Interannual variability, VT (dimensionless)

Interannual variability was calculated as the interannual coefficient of variation of fire incidence. Giglio et al. (2006) developed a similar measure with 12-month lagged autocorrelation of 5 years of monthly MODIS fire data to assess interannual variability and periodicity of fire activity at the global scale. Low values of the coefficient of variation indicate regular fire occurrence and high values represent sporadic fire occurrence (Figure 1c).

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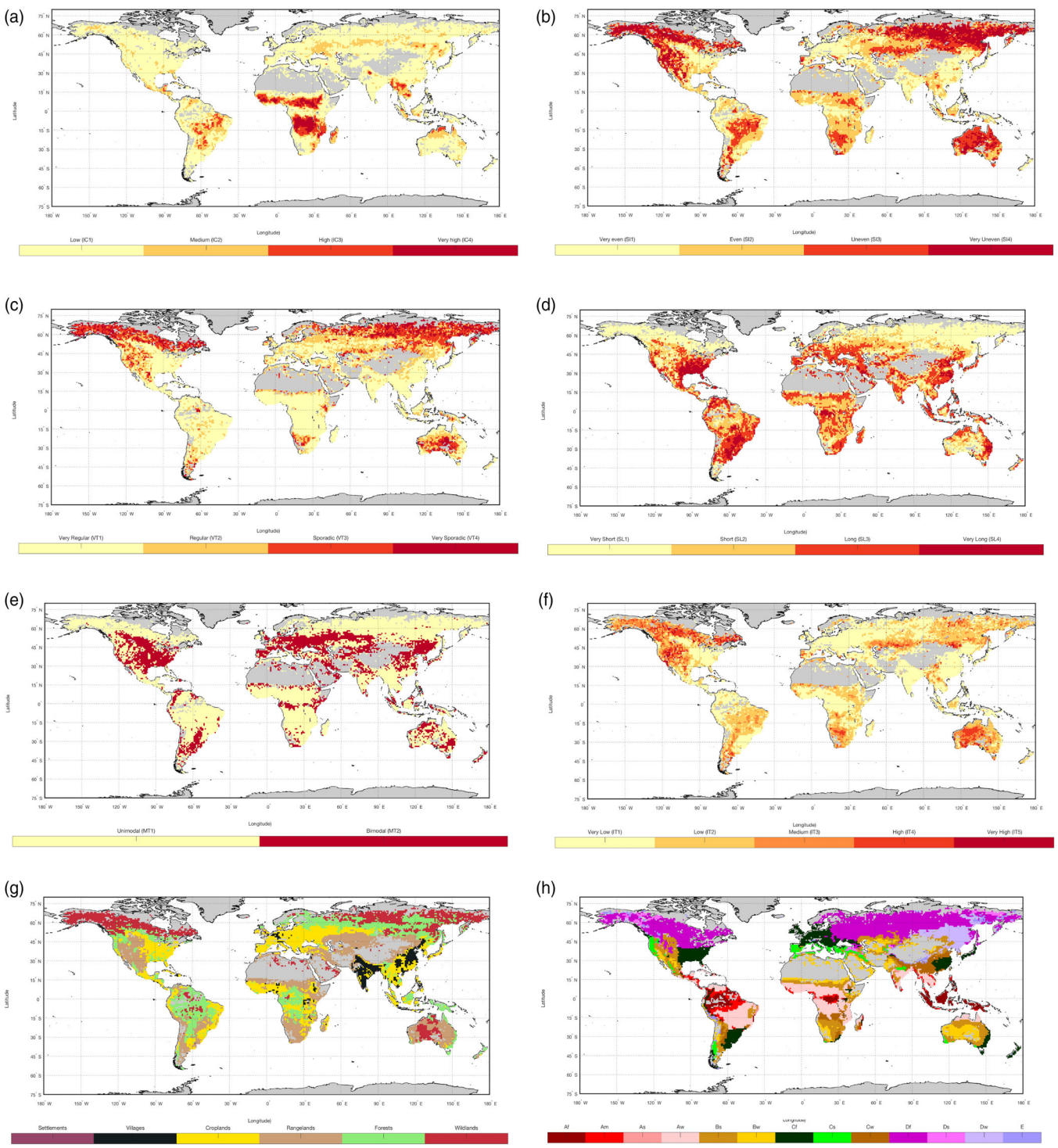


FIGURE 1 Categorized variables for global fire regime classification: (a) incidence (IC); (b) size inequality (SI); (c) interannual variability (VT); (d) season length (SL); (e) season modality (MT); (f) intensity (IT); (g) Anthromes; (h) Köppen climate types (Af, tropical rainforest; Am, tropical monsoon; As, tropical savanna, dry summer; Aw, tropical savanna, dry winter; BS, steppe; BW, desert; Cf, temperate, without dry season; Cs, temperate, with dry summer; Cw, temperate, with dry winter; Df, cold, without dry season; Ds, cold, with dry summer; Dw, cold, with dry winter; E, polar)

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Season length, SL (months)

Season length characterizes the average duration of the fire season. For each fitted von Mises distribution, the 5th and 95th percentiles of active fire counts were used to define the start and the end dates of the fire season, and its duration was calculated as the date of the 95th percentile minus that of the 5th percentile. In bimodal areas, the season length reflects the sum the durations of both seasons. Calculations were performed separately for each year and the mean fire season length was calculated considering the entire study period (Figure 1d).

Season modality, MT (dimensionless)

Season modality classifies grid cells as having one versus two fire activity peaks per year. The active fire data were composited into 16-day time periods and the shape of the active fire time series was modeled using the von Mises distribution, the equivalent of the normal distribution in circular statistics (Fisher, 1995). The number of fitted von Mises distributions varied between one and two, corresponding to uni- and bimodal distributions, respectively (Figure 1e).

Intensity, IT (MW pixel^{-1})

Intensity is measured as the 90th percentile values of fire radiative power (FRP) for the 15 years. It quantifies the rate of energy released in the vegetation combustion process at the time of the satellite observation (Wooster & Zhang, 2004). Fire radiative power is a proxy for the aerosol emission rate (Ichoku & Kaufman, 2005) and for the fire line intensity (Smith & Wooster, 2005). The FRP distribution is highly skewed towards low values, with a heavy right tail (Oliveira et al., 2015). Since very intense fires do not occur in all land cover types (Wooster & Zhang, 2004), the 90th percentile of FRP is more effective than measures of central tendency to discriminate between fire regimes (Figure 1f).

Anthromes, AT (nominal)

Anthromes, an antecedent qualitative variable, maps the Anthropogenic Biomes of the World (anthromes, v1.0) data set (Ellis et al., 2010), aggregated into six classes: dense settlements, villages, croplands, rangelands, forested lands, and wildlands (Figure 1g). Anthromes are global-scale ecological patterns resulting from long-term anthropogenic alteration of ecosystems (Ellis & Ramankutty, 2008). The anthrome data are available at 5 minutes (~9 km) spatial resolution, and combine information on population density, land cover, and land use. Version 1 of the anthrome data set (available from https://ecotope.org/anthromes/v1/data/) was preferred over version 2 because it is more directly linked with empirical data, and thus more objective, while version 2 is more appropriate for change detection studies (http://ecotope.org/anthromes/faq/#What_is_the_difference_between_Anthromes_versions_1_and_2). Anthrome data have been used in studies of anthropogenic influence on spatial and temporal patterns of fire activity (Bistinas et al., 2013; Millones et al., 2017; Oom et al., 2016; Pereira et al., 2015, 2019).

Köppen climate classification, CL (nominal)

The Köppen classification of global climates (Kottek et al., 2006) considers five main climate groups (tropical, arid, temperate, cold, and polar) divided into 14 climate types, based on patterns of seasonal precipitation and temperature. We disregarded the distinction between the tundra and frost types of the polar main climate group, irrelevant for pyrogeography, ending up with 13 climate types (Figure 1h). Use of climate classification data allows for its incorporation as an antecedent variable in the sensu lato analysis of fire regimes in the same format as the anthrome variable, facilitating the comparison of the roles of climatic and anthropic drivers. Dutta et al. (2016), Fonseca et al. (2017), and Ward et al. (2018) are recent examples of pyrogeographical studies that relied on Köppen climate classification data. These data are available at http://koeppen-geiger.vu-wien.ac.at/present.htm.

The set of variables selected cannot account for the influence of fire suppression on global pyrogeographic patterns, which may be important considering the extent of anthropogenic impacts on global fire regimes (Archibald, 2016; Benali et al., 2017; Le Page et al., 2010; Pereira et al., 2015, 2019). However, it is not feasible to incorporate the role of fire suppression in our analysis for two main reasons: the lack of globally consistent data on suppression effort, namely in Africa, which accounts for over two-thirds of global area burned (FAO, 2007), and the large uncertainties affecting estimates of fire suppression effectiveness at the landscape-scale (Calkin et al., 2014; Plucinski, 2019a, 2019b; Thompson et al., 2018).

Multiple correspondence analysis (MCA)

Correspondence analysis (CA) is a generalized principal component analysis (PCA) designed for qualitative data analysis. Multiple correspondence analysis (MCA) is an

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extension of CA used for analyzing patterns of relationships of several qualitative dependent variables (Benzécri, 1982). MCA can represent both data and variables as points in the same low-dimensional Euclidean space (Abdi & Valentin, 2007), thus providing interpretable visual representations of their relationship and, unlike PCA, is capable of dealing with qualitative variables. MCA can be used to analyze observations described by multi-level nominal or ordinal variables, and/or quantitative variables through their segmentation into bins. Each bin of each variable is coded as a binary variable, yielding a complete disjunctive matrix (CDM). Since this study relies both on qualitative variables (season modality, anthromes and Köppen climate classes) and on quantitative variables, the use of MCA was deemed necessary. Our original data matrix consists of 10,652 (n) cells by 6 (q) variables (five continuous variables, incidence, size inequality, season length, variability, and intensity, and one nominal variable, season modality) coded into a CDM where each row corresponds to a cell with six variables decomposed into 23 categories (p), according to the discretization scheme of Appendix S1: Table S1 and Figure S1, yielding a maximum of $17 = 23 - 6$ factorial dimensions. To correct for the oversampling bias and ensure equal importance of the features across all latitudes, each row (cell) was weighted by the cosine of latitude in the MCA procedure with the R package *FactoMineR* (Husson et al., 2020). Two other variables (Anthromes and Koppen-Geiger climate classes) were added at a later stage of the MCA to assist in the interpretation of results. Adjusted inertias (Greenacre, 1993) and square correlation ratios were calculated as measures of the proportion of the total inertia explained by each factorial axis and of the proportion of inertia of the coordinates of the individuals on each dimension explained by each categorical variable (Husson & Josse, 2014).

Hierarchical clustering and cluster labeling

We performed a hierarchical agglomerative cluster analysis on the cloud of observations projected onto the MCA principal factorial dimensions, applying Ward's aggregation method with the standard Euclidean distance. The use of MCA as a preprocessing step for the cluster analysis is a very common procedure, as it allows transforming qualitative data into quantitative data and contributes to improving clustering robustness by denoising the data (Kassambara, 2017). Ward's method is regarded as the most appropriate aggregation method to apply in conjunction with MCA. The optimal number of clusters was estimated as the most voted by a universe of over 20 statistical indices calculated with the R package NbClust (Charrad et al., 2014). A fine-tuning procedure was applied to the results of the cluster analysis in order to consolidate and improve group homogeneity, and to identify misclassified cells, as described in Appendix S2: Figure S1.

To derive objective labels for the clusters, we employed the v-test, a statistical test that relies on the comparison between the proportion of the number of observations of each category of each variable within a cluster, with the proportion of the same category occurring in the entire population (Lebart et al., 1995; Morineau, 1984). The higher the positive (lower the negative) value of the v-test, the more the category is over- (under-) represented in the cluster. Within each cluster, the significant levels of each fire variable (at a significance level of 1%) were sorted by decreasing order of relevance, that is, by decreasing value of their v-test scores, and the most important category of each variable retained to describe the cluster. We followed the same procedure for the levels of the antecedent variables, anthromes and Köppen climate types. However, since there are only two antecedent variables, we kept the most and second-most relevant anthrome and Köppen classes of each cluster.

Finally, we employed a version of the v-test statistic for quantitative variables to assess the degree of association between each category of an antecedent variable and each synthetic variable defined by coordinates of the individuals in each principal factorial dimension. High absolute values of the v-test indicate a strong association between the variable category and the factorial axis.

RESULTS

Multiple correspondence analysis

The MCA projected the original data onto 17 axes, with a large percentage of the adjusted inertia explained by the first two axes (68.7%, 14.3%), jointly accounting for 83% of the total adjusted inertia, while the inclusion of additional axes only marginally improved the adjusted inertia. Hence, only the plane of the two principal factorial dimensions, hereafter referred to as principal factorial plane (PFP), was kept for the subsequent analyses.

Computation of the square correlation ratios (Appendix S3: Figure S1) revealed that the VT variable, with a high value on the first axis and a very low value on the second axis, is mostly linked with the principal dimension of variability, in agreement with the geometrical arrangement of the levels of this variable along the first axis in the PFP (Figure 3). A similar conclusion

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holds with respect to the second dimension of variability and the IC variable. Thus, and for the sake of simplicity, we will refer to these two axes by the abbreviations of the interannual variability (VT) and incidence (IC) variables, respectively. Appendix S3: Figure S1 displays the degree of association between each variable and each of the two principal factorial dimensions, quantified by their square correlation ratios.

Hierarchical clustering and sensu stricto fire regime characterization

Figure 2 shows the dendrogram produced by the cluster analysis applied to the scores of the individuals on the two principal factorial dimensions. The optimal number of three clusters, hereafter designated by fire macroregimes, corresponds to the most voted, by a large majority, within the range between 2 and 10. A second hierarchical level was defined to allow better discrimination within each macroregime, considering the most voted number of clusters in the range between 5 and 10 (since four clusters did not yield a significant improvement over three clusters). An optimal number of six clusters was obtained, splitting each fire macroregime into two fire regimes, one prototypical and the other transitional. The prototypical fire regimes are designated 1a, 2a, and 3a and the transitional regimes 1b, 2b, and 3b (Figure 2).

The cluster analysis was followed by a consolidation procedure applied internally to each fire macroregime, to improve homogeneity of its prototypical and transitional regimes (Appendix S2: Figures S1 and S2). The resulting fire regimes are outlined by their convex hulls in Figure 3, together with a modified dendrogram showing the most important variables associated with each split (according to the axis involved). Each of the six sensu stricto global fire regimes, depicted as a terminal node of the dendrogram, is characterized by the most important category of each variable for the regime, according to the *v*-test scores (Appendix S4: Table S1). The first split of the dendrogram occurs along the first axis and places the two regimes of fire macroregime 1 on the positive side of the VT axis, with a fire regime 1a characterized by a very short season, very sporadic burning and very uneven fire sizes. Fire regime 1b also displays very short seasons, but more regular burning and less uneven fires sizes than 1a. The other branch of this split leads to macroregimes 2 and 3, located on the negative side of the VT axis and

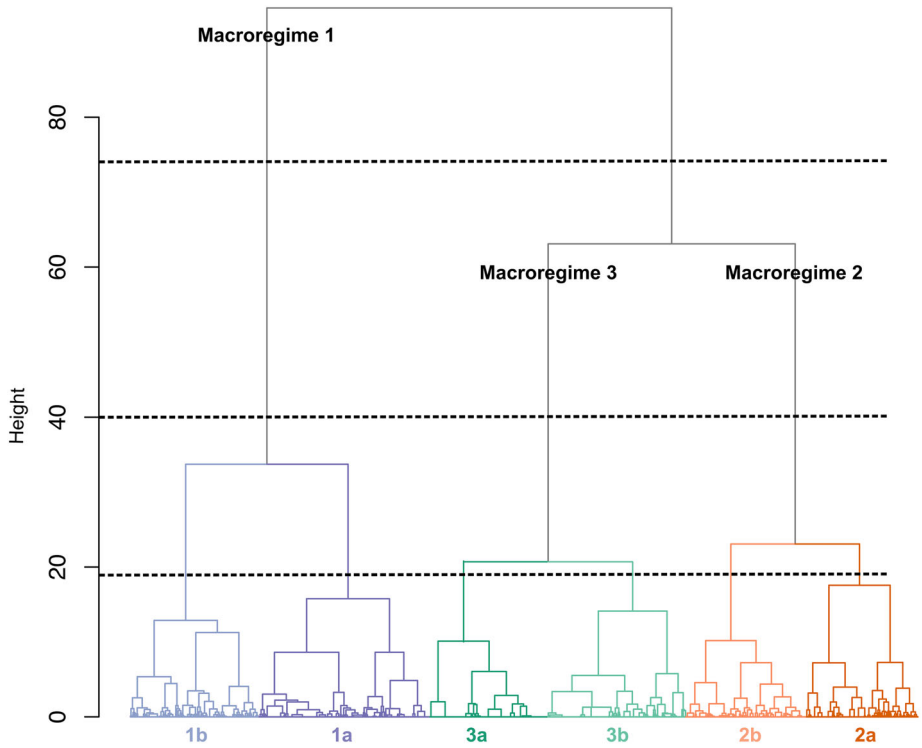


FIGURE 2 Dendrogram of the hierarchical clustering method applied to the scores of the individuals projected on the MCA principal factorial plane. The height at which two vertical branches are merged together is related to the increase in the overall intra-cluster variability due to the aggregation of the two corresponding clusters

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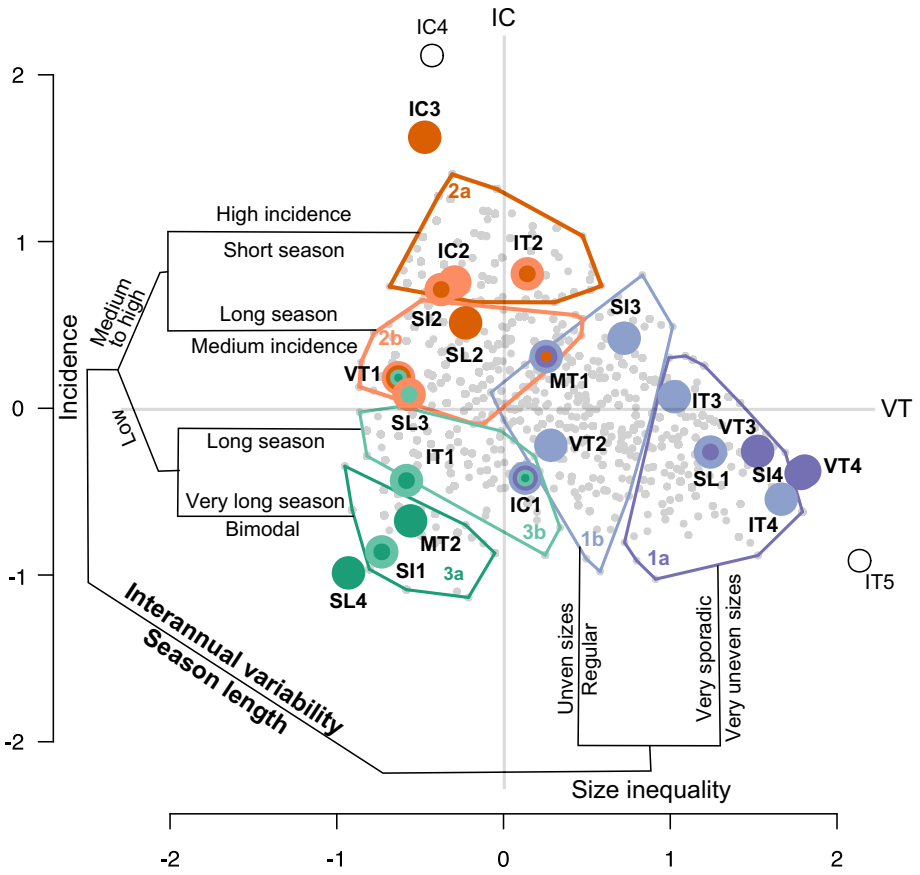


FIGURE 3 The six clusters and 23 categories associated with the sensu stricto fire variables in the principal factorial plane. Observations are depicted using small gray dots and are superimposed when characterized by the same categories for all variables. Clusters are outlined by the convex envelopes obtained after the consolidation procedure described in Appendix S2. The modified dendrogram represents the hierarchical structure defined by the pairwise distances between the clusters. Categories are depicted using large colored dots and encode the membership in the respective regime, based on *v*-test scores. When a category belongs to more than one regime, the corresponding dot is shown in the colors associated with the respective regimes. The main splitting variables are in bold and associated variable categories in normal font. Categories correspond to sensu stricto variables incidence (IC), size inequality (SI), interannual variability (VT), season length (SL), season modality (MT), and intensity (IT). Appendix S1: Table S1 shows the ordinal/nominal labels for the respective category

unfolding along the IC axis. Both regimes of macroregime 2 have very regular burning, but fire incidence is much higher and season length somewhat shorter in regime 2a than in regime 2b, but longer than in macroregime 1. The main features of fire macroregime 3 are very even fire sizes, very low intensity, and very regular burning. Regime 3a is bimodal with very long fire season, while regime 3b has a long fire season and undefined modality (Appendix S5: Figure S1) shows radar plots of the six fire variables for each fire regime, highlighting the differences and similarities between and within macroregimes). Arrangement of the fire regimes on the MCA PFP reveals that 1a, 2a, and 3a form the vertices of a triangle and are characterized by more extreme categories of the fire variables than regimes 1b, 2b, and 3b. The latter tend to display intermediate levels of the categories and occupy mid-range positions along the VT and IC.

Sensu lato fire regime characterization: Anthromes and climate types

Figure 4 shows the anthrome and Köppen climate class data (antecedent variables) projected onto the MCA PFP, highlighting the relationships between the fire regimes defined solely in terms of fire variables (i.e., sensu stricto outlined by their convex envelopes) and the antecedent variables, for a sensu lato fire regime characterization. Regime 1a is typical of wildlands in cold climates (Köppen types Df and Ds) and regime 1b occurs in cold (Df) and in steppe climate (BS) rangelands and forests. Regime 2a is found in croplands and forests of tropical and temperate winter dry climates (Aw and Cw), and regime 2b in tropical seasonally dry (Aw) and steppe (BS) croplands and rangelands. Finally, regime 3a is characteristic of croplands and villages in temperate climates

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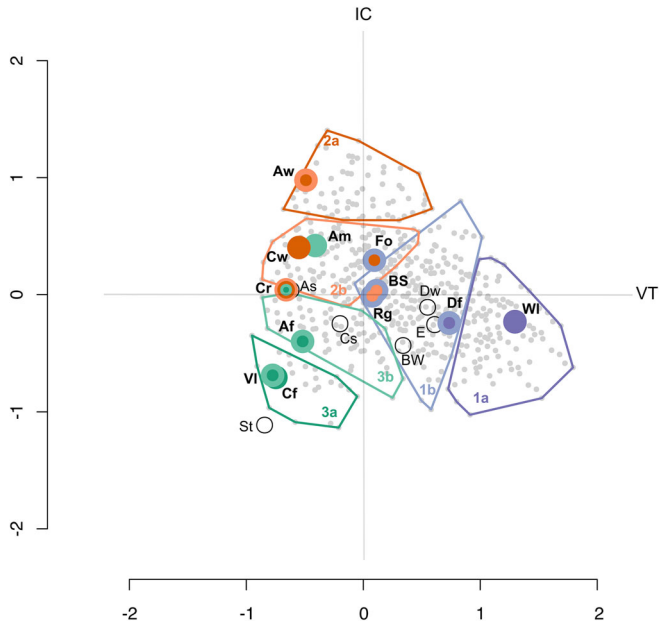


FIGURE 4 Representation on the MCA principal factorial plane of the six clusters and 19 categories associated with the antecedent variables, together with the projection of the 10,652 cells. The six clusters are represented by their convex envelopes and category colors encode their membership in the regime, based on the *v*-test score. Categories associated with the sensu lato variables anthromes and Köppen climate types are defined in Table S1 (SI). Categories in boldface text represent the two higher values of the *v*-test score for anthromes and Köppen climate types in each cluster. Three climate types are displayed in bold for the Wild-transition (Wt) cluster because the BW and Df climates have almost identical *v*-test scores

without dry season (Cf), and in tropical rainforest climates (Af), while regime 3b occurs in villages and croplands of tropical rainforest (Af) and tropical monsoon (Am) regions.

The distribution of anthromes and climate types on the MCA PFP suggests that the VT and IC axes represent gradients of the antecendent variables. The *v*-test scores for the association between categories of antecedent variables and the PFP axes (Appendix S4: Table S2), and in parenthesis in the following sentences show that the two categories most strongly associated with the VT axis are the anthromes wildlands (47.6) and croplands (−36.7). Forests (14.4), located on the wildland side, and villages (−15.4), on the cropland side of the gradient, although less strongly associated with the VT axis, strengthen its interpretation as a gradient of anthropogenic influence. Köppen climate types, also with weaker association to the VT axis, are arranged along a temperature gradient with cold climates on the positive side and tropical and temperate climates on the negative side. Climate types cold, with dry summer (Ds, 33.1) and temperate without dry season (Cf, −23.4) mark the extremes of the gradient. In contrast, the two categories with the largest *v*-test scores for association with the IC axis are climate types, namely tropical savanna with dry winter (Aw, 39.4) and temperate without dry season (Cf, −22.1), which define the ends of a rainfall seasonality gradient. The desert (BW) and tropical rainforest (Af) climates appear at almost the same position on the IC axis because both display markedly aseasonal rainfall, although with very limited rain in the former case and very abundant in the latter. Anthromes are much more weakly associated with the IC than with the VT axis. Villages and forests display the strongest associations (*v* test scores of −15.4 and 14.4, respectively).

Table 1 provides a full description of the significant sensu stricto features of each of the six fire regimes and their associated antecendent variable categories, yielding the sensu lato fire regime characterization. The rightmost column adds examples of representative ecoregions (Olson et al., 2001) for each fire regime. We propose nominal designations for the fire macroregimes: Wild fire (W) for macroregime 1, Tamed fire (T) for macroregime 2 and Domesticated fire (D) for macroregime 3. This terminology refers to the metaphor of fire as an herbivore (Bond & Keeley, 2005), borrows the designation "tamed fire" from Bowman (2003) and Pyne (2012), and reflects underlying gradients in values of both fire variables and antecedent variables. Regimes 1a, 2a, and 3a were redesignated Wild-prototypical (W_p), Tamed-prototypical (T_p), and Domesticated-prototypical (D_p). The remaining regimes 1b, 2b, and 3b were renamed Wild-transition

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TABLE 1 A sensu lato characterization of the fire (macro) regimes in terms of fire descriptors, anthromes and climate types

Fire macro regime (sensu lato)	Köppen climate types	Anthromes	Fire regime descriptors (sensu stricto)	Olson ecoregions
Wild (1) (13.6%)	Cold (Df, Ds)	Wildlands	Very sporadic to sporadic, very uneven sizes, very short season, unimodal, low incidence, intense (1a, Wild-prototype, W_p, 7.7%)	West Siberian Taïga, Eastern Canadian Shield Taïga
	Cold and steppe (Df, BS)	Rangelands, forests	Regular, uneven sizes, very short season, low incidence, medium intensity, unimodal (1b, Wild-transition, W_t, 5.9%)	Great Basin shrub steppe, Kazakh semi-desert, Patagonian steppe
Tamed (2) (74.4%)	Tropical savanna with dry winter (Aw), temperate with dry winter (Cw)	Croplands, forests	High incidence, very regular, low intensity, even sizes, unimodal, short season (2a, Tamed-prototype, T_p, 54.3%)	West Sudanian savanna, *Cerrado*, Central Indochina dry forest
	Tropical savanna with dry winter (Aw), steppe (BS)	Croplands, rangelands	Even sizes, very regular, medium incidence, long season, low intensity (2b, Tamed-transition, T_t, 20.1%)	*Caatinga*, Guinean forest-savanna mosaic
Domesticated (3) (12.0%)	Temperate without dry season (Cf), tropical rainforest (Af)	Croplands, villages	Bimodal, very regular, very even sizes, very low intensity, very long season, low incidence (3a, Domesticated-prototype, D_p, 4.5%)	Central tall grasslands, Huang He plain mixed forests, Deccan thorn scrub forests
	Tropical rainforest (Af), tropical monsoon (Am)	Villages, croplands	Very even sizes, very low intensity, very regular, long season, low incidence (3b, Domesticated-transition, D_t, 7.5%)	Central Deccan Plateau dry deciduous forests, Ucayali moist forests

Note. The fire descriptors for each fire regime are sorted by their *v*-test score. The distribution of active fires per (macro) regime is shown in percent.

(W_t), Tamed-transition (T_t), and Domesticated-transition (D_t), respectively. These designations are convenient, but acknowledgedly simplistic handles, considering the heterogeneous, dynamic nature of the socioecological systems that frame fire regimes. They do not refer to the causes of ignitions (i.e., natural vs. human) in each regime, but allude to the extent to which their fire variable and antecedent variable categories indicate diverse degrees of anthropogenic control over spatial, temporal, and behavioral fire patterns.

Global fire regimes map

Global distribution of the six fire regimes is shown in Figure 5. Macroregime Wild (fire regimes W_p and W_t) prevails at the high latitudes of the Northern hemisphere, Western United States, Central Australia deserts, and the deserts of Southwest Africa. Macroregime Tamed (fire regimes T_p and T_t) fires are mostly tropical, corresponding to savannas in Africa, South America, Southeast Asia, and Northern Australia. Macroregime Domesticated (fire regimes D_p and D_t) occurs in intensely managed lands of the Eastern United States, Europe, India, China, and temperate South America.

Areal distribution of the three macroregimes is very balanced: 30.5% (16.1% + 14.4%), 34.5% (15.0% + 19.5%) and 35.0% (16.3% + 18.7%) for macroregimes Wild ($W_p + W_t$), Tamed ($T_p + T_t$), and Domesticated ($D_p + D_t$), respectively. However, the active fire distribution is strongly dominated by the Tamed macroregime (79.0%), in

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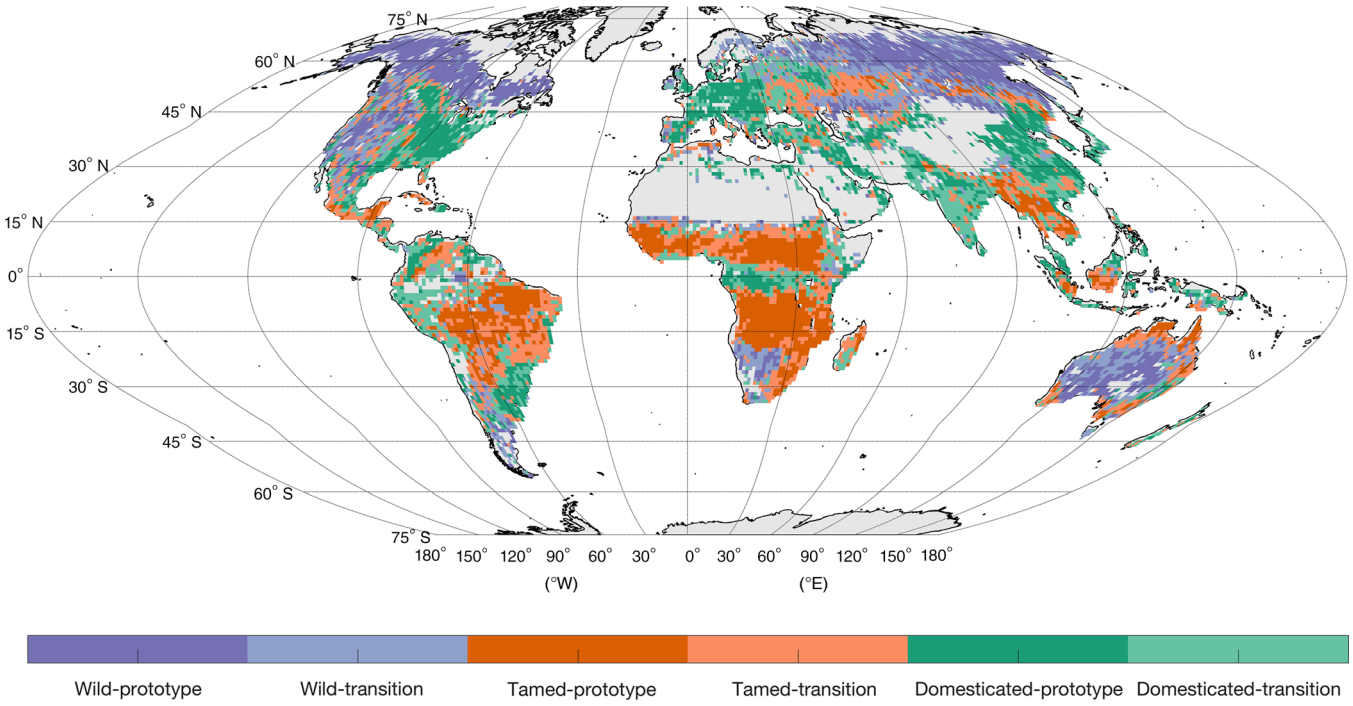


FIGURE 5 Global distribution of the six sensu stricto fire regimes identified in the dendrogram of Figure 3, labeled according to the sensu lato analysis

agreement with the high fire incidence values that characterize fire regime T_p. The remaining active fires are very evenly distributed between the Wild (9.6%) and Domesticated (11.4%) macroregimes.

Figure 6 is a two-class map of wildfire versus managed fire, where the latter class combines the Tame and Domesticated macroregimes, which jointly account for 69.5% of the global burnable area and 86.4% of the global active fire counts. The two classes represented in this map correspond to the first split in the dendrogram of Figure 2 and constitute the topmost level in our hierarchical analysis of global pyrogeography.

DISCUSSION

Hierarchical clustering and fire regime characterization

The arrangement of the six regimes and respective fire variable categories shown in Figures 3 and 4 reveals underlying gradients determined by the strength of association of the fire variables to each of the principal axes yielded by the MCA (Appendix S3: Figure S1). The horizontal axis forms a gradient primarily shaped by interannual variability (VT) in the number of active fires, which decreases from the W_p regime, where fire occurrence is very sporadic, to the managed regimes (T_p, T_t, D_p, and D_t) on the negative side of the axis, where burning is a very regular event. Fire incidence (IC) is the main determinant of the vertical axis and decreases from the high and very high values of the T_p regime, down to the low incidence of regime D_p. All other variables split their contributions more evenly between the two axes (Appendix S3: Figure S1). Therefore, global fire regimes are organized on a plane mainly defined by interannual variability in fire activity and by fire incidence, where highest fire incidence occurs at the region of lowest interannual variability. Anthromes and climate types also define clear gradients on the PFP of the fire variables. The VT axis supports a gradient of land use intensity that unfolds from the wildlands and cold climates associated with the W_p regime, to the temperate and tropical climates and croplands characteristic of the Tamed and Domesticated macroregimes. The IC axis underlies a gradient in rainfall seasonality on temperate and tropical climates, ranging from markedly seasonal for the regime T_p to aseasonal in D_p. Our hierarchical clustering approach further identified a higher level of aggregation into Managed and Wild regimes, corresponding to the first split in the clustering dendrogram (Figure 2). Figure 7 summarizes these relationships.

Wild fire: Boreal forest and deserts

Despite representing very different biomes, broad areas of boreal forests of Eurasia and North America, and of

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arid grasslands in Central Australia and SW Africa were classified under the W_p regime. Both biomes fit the model for large, infrequent fires proposed by Meyn et al. (2007). Boreal forests are biomass-rich, rarely dry ecosystems, where high fuel moisture constrains fire occurrence. Vegetation burns only when weather conditions are dry, which does not happen often. Deserts are biomass poor, typically dry ecosystems where net primary productivity is low and

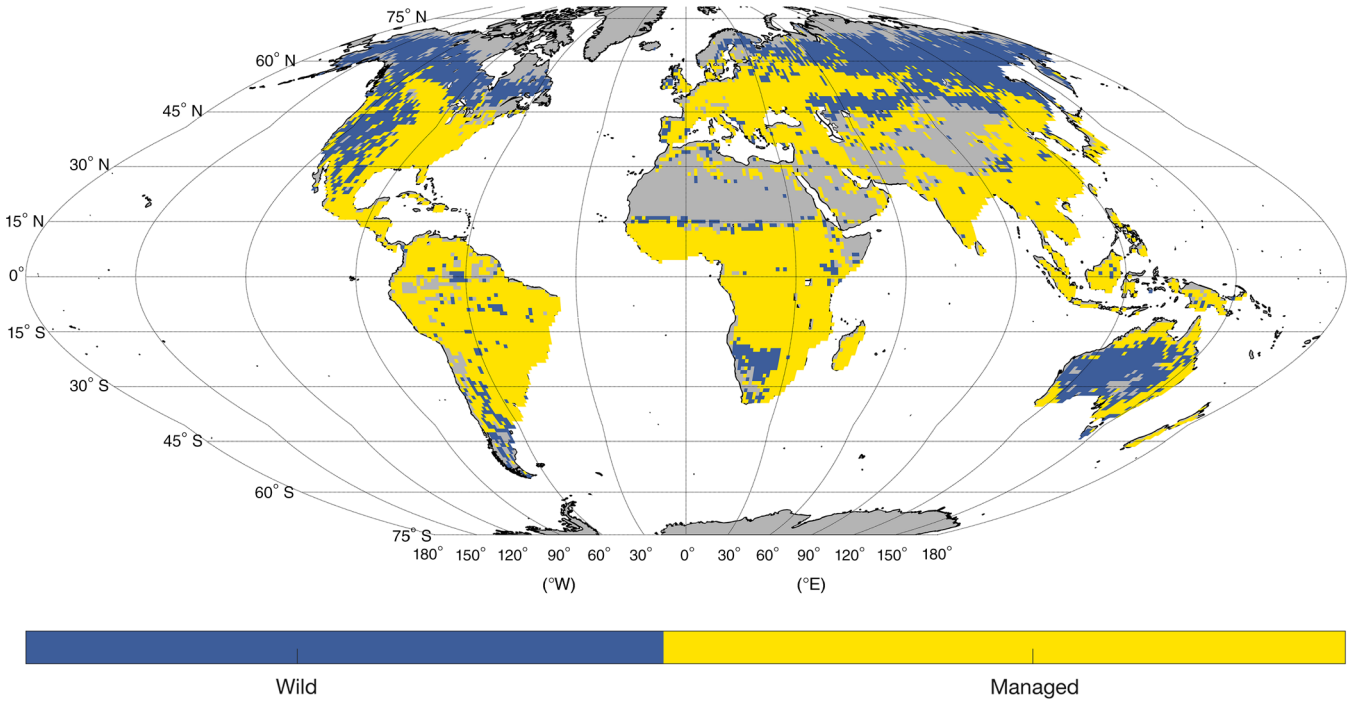


FIGURE 6 Global distribution of the Wild (W_p and W_t) and Managed (T_p, T_t, D_p, and D_t) fire regimes, corresponding to the first split in the dendrogram of Figure 2

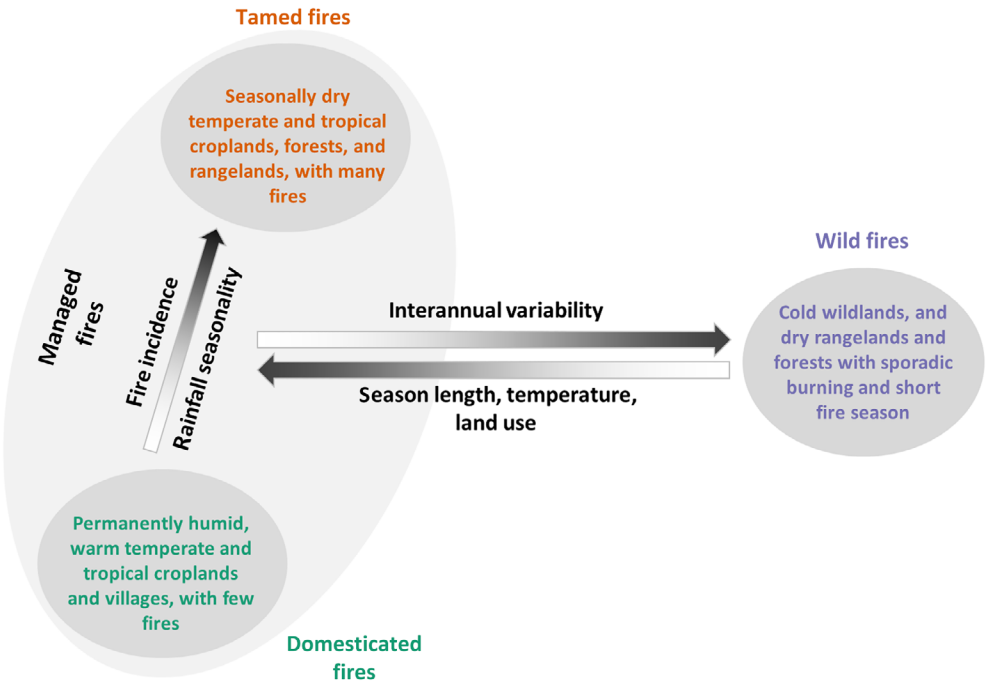


FIGURE 7 Conceptual diagram of the proposed fire regime classification, highlighting the main variables associated with the two gradients and the essential features of each fire macroregime

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fuel availability is the limiting factor for fire occurrence. These limitations were analyzed under the varying constraints hypothesis (Krawchuk and Moritz, 2011) and the global fire–productivity relationship (Pausas & Ribeiro, 2013), and explain the sporadic character (i.e., large interannual variability) of fire occurrence in boreal forests and in deserts. The typically large fire sizes of the Wild regime are due to the extensive and unfragmented character of wildland anthromes (Greenville et al., 2009; Hantson et al., 2017). Both biomes have intense fires (Figure 1f and Table 1), but typically due to distinctive fire behavior: slower spreading fires that burn large amounts of woody biomass per unit area in boreal forests and faster spreading fires burning lighter loadings of fine fuels in arid grasslands (Andrews et al., 2011). Therefore, different climatic constraints acting on different vegetation types yield similar fire regime descriptors, placing boreal forests and arid grasslands in the same sensu stricto fire regime. Archibald et al. (2013) also concluded that very distinctive biomes, for example, xeric shrublands and boreal forests, belong in the same pyrome. They attributed this result to similarity in fuel type and fuel growth constraints. Our interpretation differs in that we believe that the sensu stricto fire regime similarities of arid grasslands/shrublands and boreal forests emerge despite differences in fuel types (Pettinari & Chuvieco, 2016) and biomass accumulation constraints.

The identification of climate as the long-term driver of the occurrence of large, infrequent fires in rarely dry, biomass-rich and in sporadically wet, biomass-poor ecosystems (Meyn et al., 2007) is expressed in our analysis by the proximity between climate types desert (BW) and tropical rainforest (Af) along the IC axis of Figure 4. This axis is also a gradient of rainfall seasonality, where both BW and Af are markedly aseasonal. These climates lack the periodic alternation of seasons favorable for fuel accumulation and fuel drying that characterizes high fire incidence regions with regular burning, such as tropical savannas and Mediterranean-type ecosystems (Archibald et al., 2009; Russell-Smith et al., 2007; Saha et al., 2019).

Fire management and climate

Previous interpretations of sensu stricto descriptors of fire regimes in terms of the relative roles of humans and climate lend support to our analysis. The presence of people increases the number of fires relatively to the natural incidence due to lightning (Archibald, 2016) and can override the effect of climate (Syphard et al., 2017). Lauk and Erb (2009) reported that humans are responsible for more than 90% of the ignitions in tropical rainforests, savannas and semi-deciduous tropical forests, temperate forests and sclerophyllic woodlands. In steppe, desert, tundra, and mountainous regions anthropogenic ignitions account for more than 80% of all fires. Only in boreal forests is lightning the main source of ignition for vegetation fires.

Hantson et al. (2015) determined that the global distribution of fire sizes is more dependent on human activity than on natural factors. The most uneven fire size distributions occur in boreal forests and Australian deserts, while the most even occur in European, American, and Australian croplands. Oom et al. (2016) also identified a gradient in fire size distribution, ranging from evenly sized distributions with many small fires in used anthromes (the combination of the dense settlements, village, cropland, and rangeland anthromes; Ellis et al., 2010), to distributions dominated by a few very large fires in wildland anthromes. Archibald, Nickless, et al. (2010) and Laris (2013) both considered that the high regularity of fire activity in Africa reveals the strength of human influence on fire regimes, which supersedes the effect of interannual climatic variability. The global scale analysis by Giglio et al. (2010) showed that fire regularity is higher in the tropics, especially in African savannas, and lower in central Australia, the United States, and boreal forests of Asia and North America. Length of the fire season is extended in anthropogenic fire regimes due to diverse land management requirements (Le Page et al., 2010; Vigilante et al., 2004) and is longest in croplands of economically developed regions (Chuvieco et al., 2008). Due to anthropogenic ignitions, the U.S. fire season currently is about three times longer than that caused by lightning sources alone (Balch et al., 2017).

In our analysis, fire size inequality, interannual variability, and season length all bear more heavily on the VT axis of Appendix S3: Figure S1, contributing to define the managed fire versus wildfire gradient shown in Figure 7. In agreement with previous research, wildfires in unmanaged lands occur more sporadically, over shorter seasons, display more uneven sizes than managed fires in used anthromes, which burn very regularly in longer fire seasons, and have more evenly distributed sizes. Human intervention also affects the number of modes in the fire season. Benali et al. (2017) found that bimodal areas have higher population densities and a larger fraction of active fires in managed lands than neighboring areas with a unimodal fire season. Our results, where only the D_p fire macroregime exhibits bimodal fire seasons, corroborate their findings.

Although African savannas burn very extensively, they usually are unaffected by extreme fire events (Laris, 2013) and display an inverse relation between fire radiative power and land use intensity (Archibald, Scholes, et al., 2010). At global level, the most intense

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fires occur in deserts of Australia and Namibia, in Patagonia, and in conifer forests of boreal North America and the U.S. West, while croplands typically display low-intensity burning (Laurent et al., 2019), as shown in Figure 1f. These patterns reappear in our classification of fire regimes, where fire intensity is highest in the wildlands of the W_p regime, decreases towards the forests and croplands of the Tame macroregime, and is lowest in the villages and croplands of the D_p regime.

Managed burning prevails in Africa, South America, Eastern North America, South Asia, parts of Central Asia, and Europe (Figure 6). Wildfires predominate in the boreal forests of Eurasia and North America, the Western US, the deserts of Southwest Africa, and throughout most of the Australian hinterland. Australia is the continent with the largest proportion of area under the Wild macroregime. North America and Eurasia display balanced combinations of the two types of macroregimes. In the United States, a well-defined meridional division separates wildfires in the West from managed fires in the Midwest and Eastern parts of the country (Figure 6).

It is arguable whether the W_t regime should be considered Wild or Managed, taking into account its characteristically low interannual variability in fire activity, medium fire intensity and association with rangelands, a used anthrome (Ellis et al., 2010). In the latter case, managed regimes would account for 83.9% of the global flammable area and 95.2% of all active fires. This percentage of active fires is very close to that reported by Pereira et al. (2019) as occurring in anthromes that display weekly cycles in fire activity (93.7%), a very specific diagnostic of anthropic influence since there are no natural cycles with weekly periodicity. Even if we admit that W_t regime fires are not managed, the numeric results shown in Table 1 and in Figure 6 evidence the extent of human influence upon contemporary global pyrogeography, that is, its fully Anthropocenic character.

Fire regime changes and spatial outliers

The classification framework proposed can support the interpretation of fire regime changes resulting from modifications in ignition patterns and land use. The finding that fire seasons are getting longer in the USA due to growing human influence over fire regimes (Balch et al., 2017), is interpretable as a displacement along the main fire regime classification gradient, where season length increases with land management intensity. Grégoire et al. (2013) and Andela and Van Der Werf (2014) showed that extensive land use conversion from pastoral rangelands to croplands in African savannas is leading to a reduction in area burned. In our fire regime classification framework, this corresponds to a transition from the T_p to the D_p fire regime, along the respective anthropogenic influence gradient, and associated with a decrease in fire incidence, size, and intensity (Figure 7). The fire regime changes observed in northern Kazakhstan during the 1990s provide an example of the opposite trend. The collapse of the Soviet Union, and subsequent disruption of social and political organization, led to rural land abandonment. Extensive areas of croplands and rangelands started to revert to steppe, and flammable fuels accumulated. During that decade, the number of fires increased eightfold and area burned increased sevenfold (Dara et al., 2019). This pattern is opposite to that observed in Africa (Andela & Van Der Werf, 2014; Grégoire et al., 2013) and corresponds to a migration of the fire regime up the IC axis, towards higher fire incidence and larger, more intense fires (Figure 7). In parts of southern Europe, rural depopulation and agricultural land abandonment have led to the expansion of forests and shrublands, increasing the extent and continuity of areas with highly flammable, heavy fuel loadings (Loepfe et al., 2010; Moreira et al., 2011). The fire regime has experienced an increase in the interannual variability of area burned (Fernandes et al., 2020), and in the number and size of very larges fires, leading to more extreme fire size inequality (Royé et al., 2019). These changes represent a decrease in land use intensity and a fire regime turning feral (Bowman, 2003), that is, shifting to the right along the horizontal axis of Figure 7. Paradoxically, the increasing prevalence of very large fires and of burned area interannual variability may be aggravated by fire management policies that emphasize fire suppression to the detriment of landscape-scale fuel management, leading to the fire-fighting trap discussed by Moreira et al. (2020).

The choice of a 1° grid cell for our analysis, a spatial resolution four times lower than that used by Chuvieco et al. (2008) and by Archibald et al. (2013) did not hinder our ability to detect small spatial outliers, that is, areas with fire regimes contrasting with those of their neighbors. The patch of T_p fire regime in the Flint Hills of Kansas, US, exemplifies such a spatial outlier. It is the largest intact remnant of North American tallgrass prairie and undergoes frequent prescribed burning, to mimic the historic fire regime (Tulbure et al., 2011). Between 2000 and 2010 the annual percentage area burned in the region varied between 15% and 48% (Mohler & Goodin, 2012), substantially higher than throughout the Eastern United States, most of which is classified as Domesticated fire (Figure 6).

The white sand ecosystems of Amazonia provide another example of a small spatial outlier fire regime.

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These ecosystems include open areas and woodlands, usually located in white quartz sand soils, often waterlogged or seasonally flooded (Adeney et al., 2016). Our global fire regime map identifies a patch of W_p regime in the Brazilian state of Amazonas, north of the town of Barcelos, in a region with varied fire regimes, including D_t, W_t, and T_t, and areas without significant fire activity (Figure 5). The open canopies and high fuel loadings of thick root mat layers make white-sand ecosystems fire-prone during dry periods, resulting in low fire incidence and sporadic burning in minimally disturbed areas of the Central Amazon. Figure 1 shows that the area has large values of fire size inequality (Figure 1b) and interannual variability (Figure 1c), which are key attributes of the W_p regime.

A few grid cells in the Punjab state of northern India form another spatial outlier, where a small patch of Tamed macroregime stands out in a vast area of Domesticated macroregime (Figure 5). Vadrevu et al. (2012) analyzed eight clusters of fire activity in India and found that, due to crop harvest residue burning, the Punjab cluster had the highest number of fire counts, although fires were smaller and less intense than in some of the other clusters. The very high fire incidence leads to the inclusion of this spatial outlier in the Tamed macroregime.

Comparison with previous global fire regime analyses

Lauk and Erb (2009) and Lauk and Karl-Heinz (2016) proposed a global typology of vegetation fires, which considered most vegetation burning as anthropogenic, with the exception of Eurasian and Nearctic boreal forests, where lightning accounts for a majority of ignitions. They split anthropogenic fires primarily by size and associated them with specific biomes and land uses: large fires, in sparsely populated tropical savannas and grasslands, and small fires in tropical forests, used in shifting cultivation. Our classification of fire macroregimes into Wild, Tame, and Domesticated enriches that proposed by Lauk and Erb (2009) and Lauk and Karl-Heinz (2016).

Chuvieco et al. (2008) used 6 years of MODIS active fire data to derive only three fire regime descriptors, namely fire incidence, season length, and interannual variability, using our terminology. Cross-tabulation of the variables yielded eight fire groups. The most widespread displayed high fire incidence, short seasons, and high interannual variability, and occurred in both boreal and tropical dry and wet biomes. We find no correspondence of this fire group with any of our six fire regimes. Chuvieco et al. (2008) interpreted their fire types in terms of population, economic, and land cover variables. They found low population densities associated with short fire seasons, high interannual variability, and low fire incidence, a relationship present in our Wild macroregime, while high population densities, highly correlated with low interannual variability, also appear in our Domesticated macroregime. High land use intensity consistently resulted in low interannual variability in fire activity. However, it had varied consequences for season length and fire incidence due to modulation by environmental and socioeconomic factors.

Archibald et al. (2013) relied on 10 years of MODIS active fire and burned area data to derive five descriptors for fire size, frequency, intensity, season length, and area burned, and used cluster analysis to combine them into five "pyromes." These pyromes, labeled in terms of fire frequency, intensity, and size, bear some relations with our fire regime classification. The frequent-intense fire pyromes share some attributes to our tamed fire macroregime, and the rare fire pyromes, both in the intense-large and cool-small variants have similarities with our W_p and W_t fire regimes. Our Domesticated fire macroregime partially overlaps their infrequent-cool-small pyrome, geographically and in terms of defining fire regime variables.

Despite these similarities, there are some important geographical differences between the pyromes map of Archibald et al. (2013) and our map of the six fire regimes (Figure 5). Some of these differences stem from the fact that we map a substantially larger area, mostly due to a longer time series, larger cell size, and because of differences in the treatment of classification uncertainty described in Appendix S2. Other differences are a consequence of actual discrepancies in regions mapped in both studies.

CONCLUSIONS

Remote sensing has played a key role in improving the accuracy of quantitative estimates of fire-affected areas and biomass burned. However, it cannot be directly used to distinguish between natural and anthropogenic fires, since we cannot observe land use and human pyrogenic activities. Nevertheless, and building upon previous research, we relied on remotely sensed data to produce information on spatial, temporal, and behavioral patterns of fire activity. This information was then converted into knowledge regarding the location, extent, and characteristics of fire regimes, aligned along gradients of human versus climatic influence.

Interpretation of the six fire regimes in terms of climatic and anthropic antecedent variables showed that anthropogenic influence and temperature line up with the interannual variability axis, such that sporadic fires are associated with wild and cold regions, while regular

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burning is typical of managed fire regimes in temperate and tropical climates. This axis is more strongly associated with anthromes than with climate types. The Tamed and the Domesticated fire macroregimes account for 90.4% of all active fires in our data set and for 69.5% of the global combustible area. Most fire activity occurs in the cropland, forest, and rangeland anthromes of tropical and temperate seasonally dry climates. Less than 10% of all active fires were found in cold wildlands, and in the rangelands and forests of steppe and desert climates of the Wild macroregime. These results highlight the dominant role of the human dimension in Anthropocene global pyrogeography.

The proposed classification scheme produced two hierarchically related global maps of fire regimes that are consistent with major pyrogeographical patterns, most notably the identification of distinctive fire macroregimes corresponding to tropical savannas, boreal forests, and the major agricultural regions of the world. The apparently incoherent inclusion of boreal forests and deserts in the same macroregime is interpretable in terms of established pyrogeographical and macroecological concepts. It is also important to acknowledge that there are interactions between the effects of climate and human activity on fire regimes, since anthropogenic greenhouse gas emissions affect climatic features at global scale, and land use patterns, including human pyrogenic activity are constrained by climate. Nevertheless, our analysis succeeded at identifying distinctive sensu stricto fire regimes, at interpreting them in terms of underlying climatic and anthropic gradients, and at showing different ways in which the role of climate is modulated by human action.

The global scale of this analysis implies that its results are not particularly applicable for national or regional fire management, and no organization has the jurisdiction to manage fire at the planetary level. Nevertheless, and despite its global scope and static nature, the proposed fire regime conceptualization and classification proved appropriate to interpret fire regime changes in regional settings that are very diverse in terms of environmental and socioeconomic conditions and dynamics, and allowed the identification and characterization of a few instances of locally distinctive fire regimes. It also contributed to clarify the relative roles of environmental factors and anthropogenic factors as fire regime drivers, and to synthesize knowledge on global pyrogeography, all of which have been identified as important motivations for fire regime mapping.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sets utilized for this research are as follows: MCD14ML collection 6 active fire data from NASA's Fire Information for Resource Management System (FIRMS) (https://earthdata.nasa.gov/firms), part of NASA's Earth Observing System Data and Information System (EOSDIS) (doi: 10.5067/FIRMS/MODIS/MCD14ML).

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